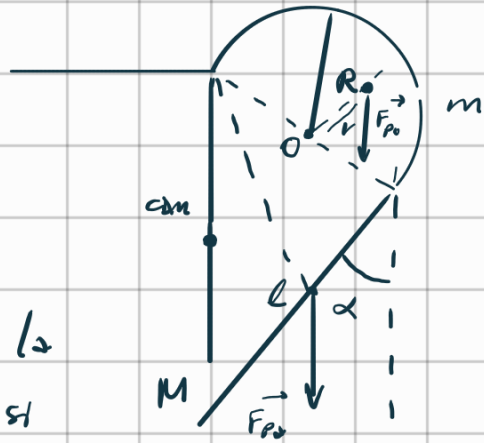


$$l = 4R$$

$$M = 5m$$

$$r = \frac{2}{\pi} R \rightarrow r = \text{distanza cm ombrello}$$



le uniche 2 forze che ci sono sono la forza peso e la normale che però si bilanciano, quindi ci interessa l'equilibrio dei momenti

$$\vec{M} = \vec{r} \times \vec{F} = \begin{cases} 1) \text{ se } \vec{F} = 0 \\ 2) \text{ se } \vec{r} \parallel \vec{F} \end{cases}$$

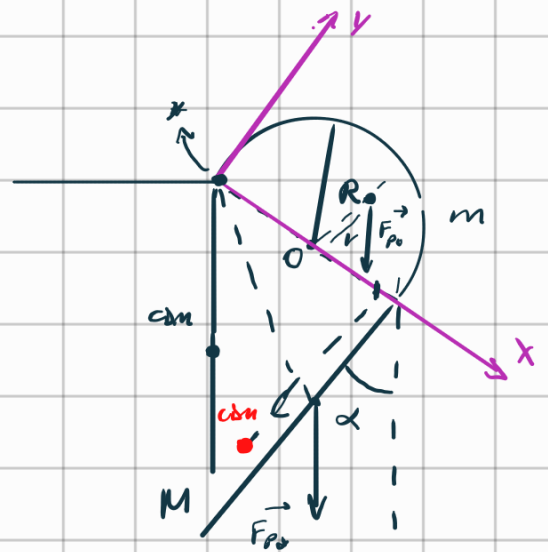
trovo il "nuovo" cm

$$\vec{r}_{cm} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i} = \frac{m \vec{r}_{cm, omb.} + M \vec{r}_{cm, st.}}{m + M} = \frac{m (x_{cm}^0, y_{cm}^0) + M (x_{cm}^1, y_{cm}^1)}{m + M}$$

$$= \frac{(m x_{cm}^0 + M x_{cm}^1, m y_{cm}^0 + M y_{cm}^1)}{m + M}$$

$$x_{cm}^0 = R \quad y_{cm}^0 = \frac{2}{\pi} R$$

$$x_{cm}^1 = 2R \quad y_{cm}^1 = -2R$$



\* Solitamente conviene mettere l'origine del SDR nel punto di vincolo

$$\vec{r}_{cm} = \frac{(11R + 10mR, 10 \frac{2}{\pi} R + 10mR)}{6m} = \left( \frac{11}{6} R, \frac{1}{3} R \left( \frac{1}{\pi} - 5 \right) \right)$$

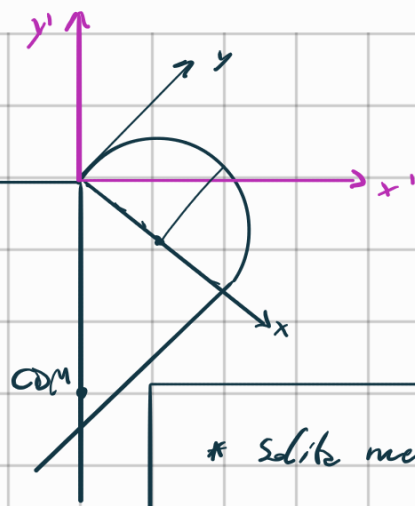
$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{cases} x' = x \cos \alpha + y \sin \alpha \\ y' = -x \sin \alpha + y \cos \alpha \end{cases}$$

$$\text{CM} \left( \frac{11}{6} R; \frac{1}{3} R \left( \frac{1}{R} - 5 \right) \right)$$

$$\vec{r}_{\text{CM}} = \frac{11}{6} R \frac{\cos \alpha}{\cos \alpha} - \frac{1}{3} R \left( 5 - \frac{1}{R} \right) \frac{\sin \alpha}{\cos \alpha}$$

$$\alpha = \arctan \left[ \frac{11}{2 \left( 5 - \frac{1}{R} \right)} \right] \approx 50^\circ$$



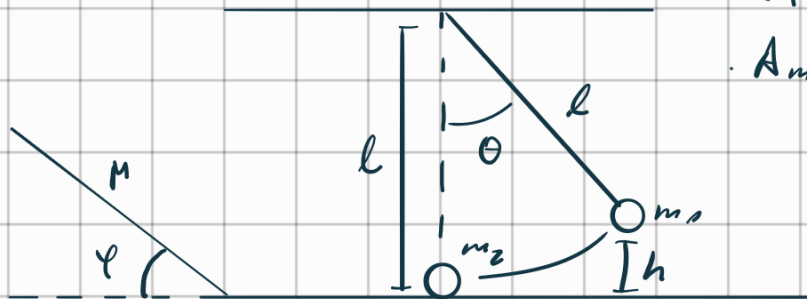
\* Solito mente è negativo, ma cambiamo segno visto che stiamo ruotando in senso antiorario

$$m_2 = 1 \text{ Kg} = m_1 \rightarrow m$$

$$l = 1 \text{ m}$$

$$\theta = \frac{\pi}{3} \quad \varphi = \frac{\pi}{4}$$

$$\mu = \frac{1}{\sqrt{3}}$$



$$h(m_2) = ?$$

$$A_{mp} = ?$$

URTO PERFETTAMENTE ELASTICO

- CONS QDM

- CONS K

$$E_i = E_f \rightarrow U_i = K_f \rightarrow mgh = \frac{1}{2} m V_0^2$$

$$gl(1 - \cos \theta) = \frac{1}{2} V_0^2 \quad V_0 = \sqrt{2gl(1 - \cos \theta)}$$

$$K_{1i} = K_{2f} + K_{1f} \rightarrow$$

$$\hookrightarrow \frac{1}{2} m V_0^2 = \frac{1}{2} m V_{2f}^2 + \frac{1}{2} m V_{1f}^2$$

$$\hookrightarrow V_0 = V_{2f} + V_{1f} \rightarrow \text{cons } K$$

$$\bullet q_i = q_f \rightarrow m V_0 = m V_{2f} + m V_{1f} \rightarrow V_0 = V_{2f} + V_{1f}$$

$$(V_{2f} + V_{1f})^2 = V_{1f}^2 + V_{2f}^2$$

$$\cancel{V_{1f}^2} + 2 V_{2f} V_{1f} + \cancel{V_{1f}^2} = \cancel{V_{1f}^2} + \cancel{V_{2f}^2}$$

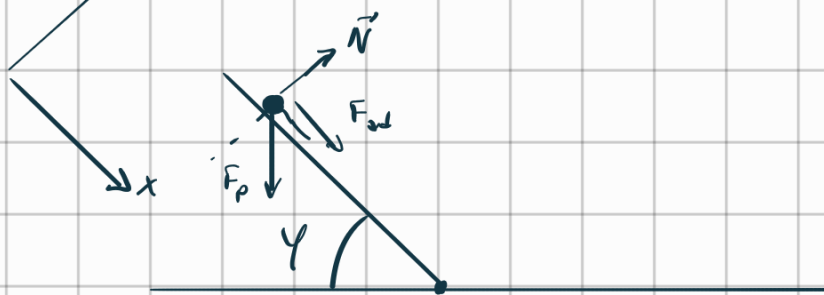
$$2 V_{2f} V_{1f} = 0 \rightarrow V_{1f} = 0 \text{ (unica soluzione valida nel nostro caso)}$$

$$V_0 = V_{2f} \rightarrow \text{post urto} \rightarrow \text{velocità con cui parte } m_2$$

$$E_i = E_f + L_{\text{attr.}}$$

$$E_f = mgh$$

$$L_{\text{attr.}} = F_A \cdot s \rightarrow F_A \cdot \frac{h}{\sin \varphi}$$



$$N - F_{py} = 0$$

$$N = F_{py} = mg \cos \varphi$$

$$F_{ad} = \mu \cdot N = \mu \cdot mg \cos \varphi$$

$$L_{\text{attr.}} = \frac{\mu m g h \cos \varphi}{\sin \varphi}$$

$$V_i = V_0 \cos \varphi$$



$$E_i = \frac{1}{2} m V_i^2 = \frac{1}{2} m V_0^2 \cos^2 \varphi$$

$$E_i = E_f + L_{\text{attr}}$$

$$\hookrightarrow \frac{1}{2} m V_0^2 \cos^2 \varphi = \underline{mgh} + \mu m g h \frac{\cos \varphi}{\sin \varphi}$$

$$\frac{1}{2} V_0^2 \cos^2 \varphi = gh + \mu gh$$

$$h = \frac{\frac{1}{2} V_0^2 \cos^2 \varphi}{(g + \mu g)}$$

\* ampiezza  $\rightarrow$  ritorno alla configurazione iniziale

